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24 February 1965
SUPERSEDING
FA(FC)-PD-99
10 July 1961

MILITARY SPECIFICATION

WRITE AMPLIFIER: 10526585

1. SCOPE

1.1 This specification covers one type of write amplifier used to write information on the memory disc of the Gun Direction Computer: M18.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F10526585	Assembly, Write Amplifier

PACKAGING DATA SHEET

M14002	Data Sheet, Master Packaging

FSC 1220

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(Copies of specification, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication.- The write amplifier shall be manufactured in accordance with Drawing F10526858 and drawings pertaining thereto.

3.2 General specifications.- The write amplifier shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The write amplifier shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the write amplifier for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the write amplifier shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The write amplifier shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The write amplifier shall meet the requirements of 3.4.1.3 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The write amplifier shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

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3.3.4 Humidity.

3.3.4.1 Operating.- The write amplifier shall meet the requirements of 3.4.1.3 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of +135°F.

3.3.4.2 Non-operating.- The write amplifier shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The write amplifier shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The write amplifier shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 0.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Performance.- Unless otherwise specified the write amplifier shall meet the performance requirements of this specification at standard ambient temperatures between 60° and 90°F and at a barometric pressure of 28 to 32 inches of mercury with a RH up to 90%.

3.4.1 Write amplifier I (see 6.1).- The write amplifier shall meet the requirements of 3.4.1.1 and 3.4.1.2 when the appropriate voltages are applied to the respective pin designations below:

PIN	VOLTAGES
15B	-1.5
51B	+1.25
31A	-6
41A	+6
31B	+35
36A	-10
29A	-18
8A	+15
32B, 23A	+15
33A	Ground (Common ground to all dc voltages)

3.4.1.1 True state (static conditions).- When the write amplifier is self-gated (per Figure 1), and in the true state, the following conditions shall exist:

(a) The saturation voltage of the "on" flip-flop transistor shall not exceed 0.3 volt (v).

(b) The back-bias base-to-emitter voltage of the "off" flip-flop transistor shall be at least 0.4 v.

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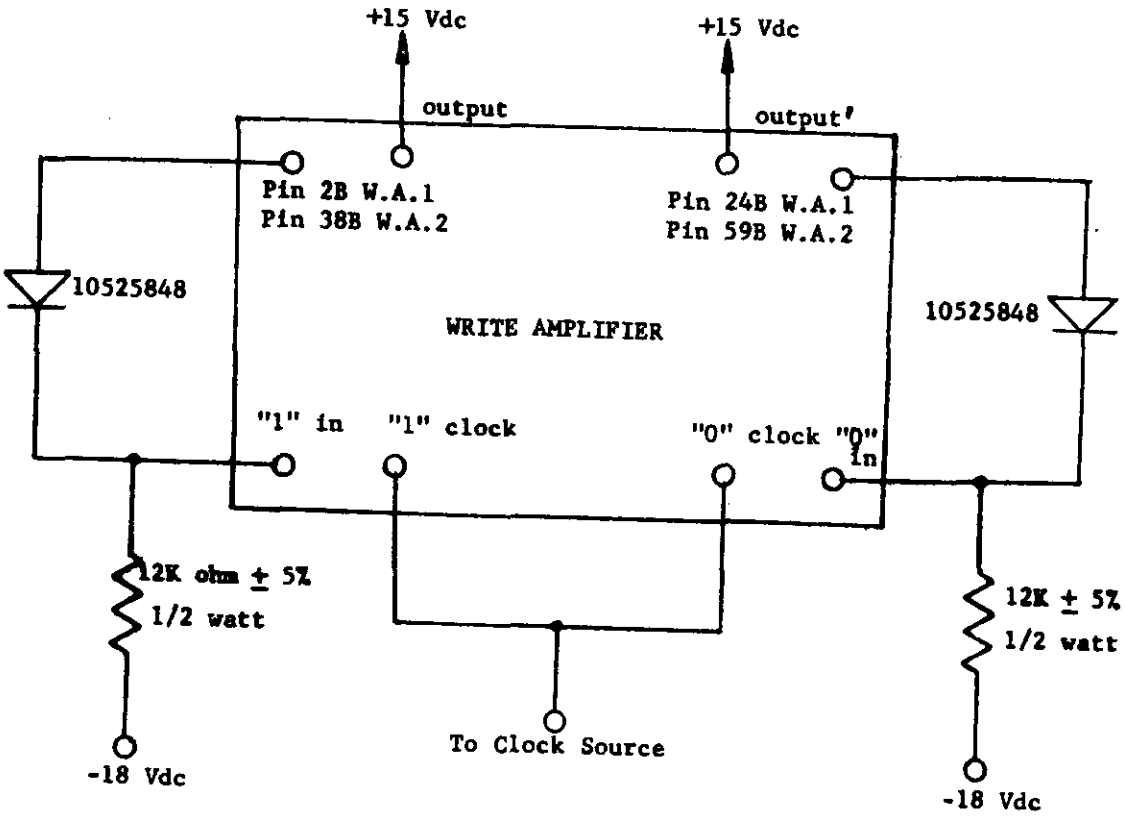


FIGURE 1 - Write Ampl fier Self-Gating System

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(c) The voltage at the base of the "on" drivers shall be within plus or minus 0.75V from ground.

(d) The voltage at the collectors of the "on" drivers shall be no more positive than plus 12 volts (V).

(e) The voltage at the collector of the "on" buffer transistor shall be minus 7.0 plus or minus 1.0V.

(f) The voltage at the base of the "off" drivers shall be minus 6.0 plus or minus 1.0V.

(g) The current through the output line shall be 64 plus or minus 2 milliamperes (ma).

(h) The current through the output prime (') line shall be no greater than 10 microamperes (ua).

3.4.1.2 False state (static conditions).- All conditions existing in 3.4.1.1 shall be analogous to the components of the same amplifier (i.e., the opposite half of the flip-flop), when the amplifier is self-gated (per Figure 1) and in the false state. (For example, the "off" transistors in the true state shall now be "on" and all conditions existing for the "on" transistors in the true state shall apply to the now existent "on" transistors in the false state).

3.4.1.3 Dynamic conditions.- The following conditions shall be met with pins 29B and 34B shorted, and the clock signal applied to either pin.

3.4.1.3.1 Minimum clock pulse.

3.4.1.3.1.1 Triggered by 3.8 - V clock signal.- When the write amplifier is self-gated (Figure 1) and triggered by a minus 3.8 -V clock signal, the waveforms at the drivers emitters shall be squarewaves varying between -0.5 ± 0.75V to -6.0 ± 0.5V.

3.4.1.3.1.2 Triggered by 1.5 -V clock signal.- When the write amplifier is self-gated (Figure 1) and triggered by a minus 1.5 -V clock signal, the levels at the driver emitter shall not change in response to clock triggering.

3.4.1.3.2 Delay time.- When the write amplifier is self-gated (Figure 1) and triggered by a 6-V clock signal, so that, drivers Q1 and Q2 (Drawing F10526588) are on, the clock signal compared to the waveform at the emitter of transistor Q1 shall exhibit the following:

(a) The waveform at the driver's emitter shall be a squarewave varying between -0.5 ± 0.75V to -6.0 ± 0.5V.

(b) The delay time, t_1 (Figure 2), measured from the time a clock pulse reaches 10 percent of its total excursion (going from minus 6 to minus 0.15V) until the waveform at the emitter of the transistor reaches 90 percent of its total excursion (going from minus 0.5 to minus 6V) shall be less than 0.4 microsecond (u sec).

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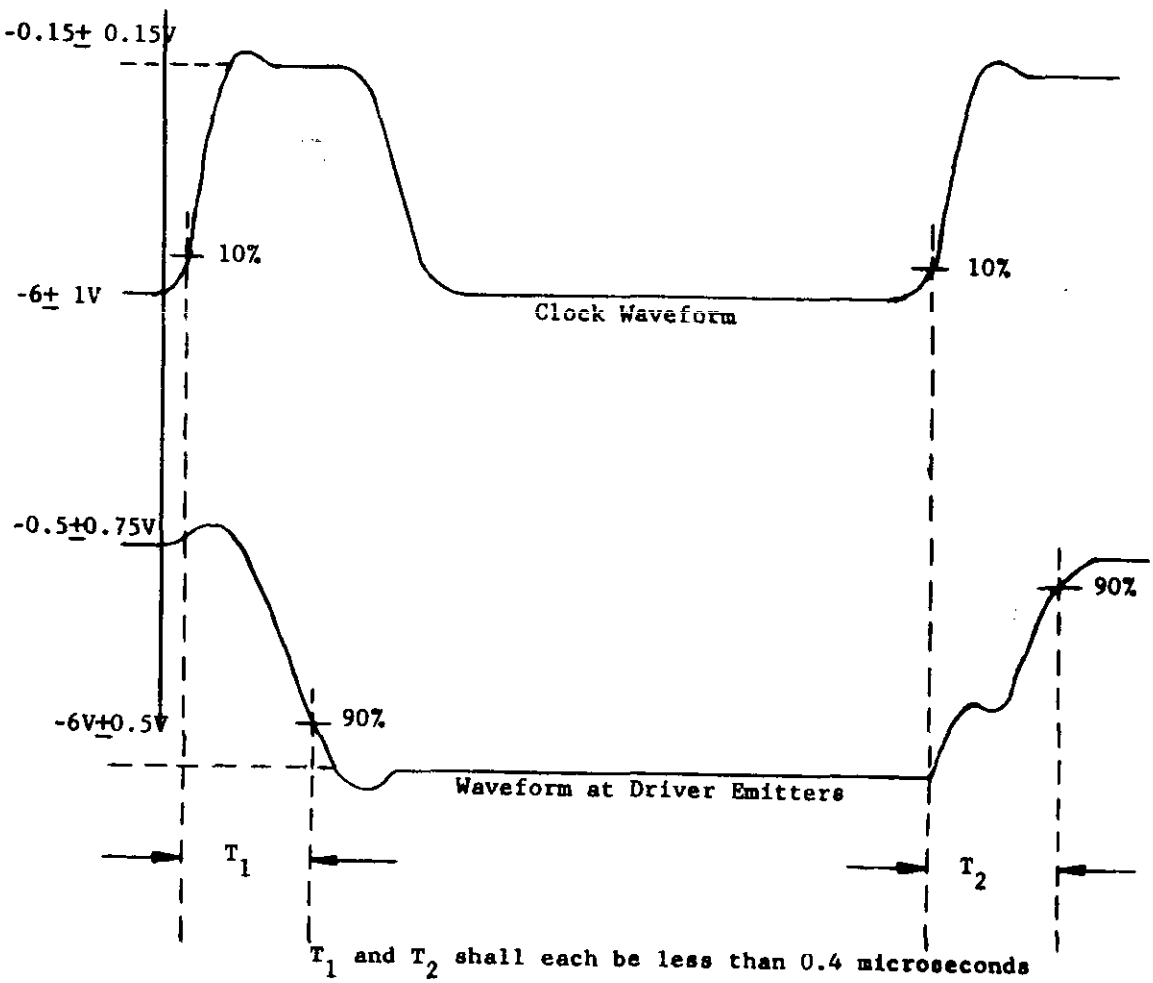


FIGURE 2 - Write Amplifier - Dynamic Waveforms

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(c) The delay time, t_2 (Figure 2), measured from the time a clock pulse reaches 10 percent of its total excursion (going from minus 6 to minus 0.15V) until the waveform at the emitter of the transistor reaches 90 percent of its total excursion (going from minus 6 to minus 0.5V) shall be less than 0.4 u sec.

3.4.1.3.2.1 With the write amplifier self-gated (Figure 1) and triggered by a clock signal, so that, drivers Q7 and Q8 are on, the clock signal when compared to the waveform at the emitter of transistor Q7 shall meet the requirements of 3.4.1.3.2.

3.4.2 Write Amplifier II (see 6.1).- The requirements of 3.4.1 inclusive shall be met, when the appropriate conditions of Write Amplifier I are made analogous to Write Amplifier II.

3.4.3 Head isolation diodes (CR42, CR43, CR44, CR46, CR47, CR48, CR49, CR50, CR52, CR54, CR57, CR61, CR62, CR64, CR66 and CR69).

3.4.3.1 Forward voltage.- The forward voltage drop across each head isolation diodes shall not exceed 1V at 90 ma.

3.4.3.2 Leakage current.- The leakage current through each head isolation diodes shall not exceed 10 ua when back-biased by 15V.

3.4.4 Test diodes (CR1, CR7, CR8, CR24, CR25, CR26, CR27, CR28, CR30, CR32, CR33, CR34, CR35, CR60, CR62 and CR68).

3.4.4.1 Forward voltage drop.- The forward voltage drop across the diodes shall not exceed 0.6V at 1.5 ma.

3.4.4.2 Leakage current.- The leakage current through the diodes shall not exceed 20 ua when back-biased by 15V.

3.4.5 Test diodes (CR5, CR29, CR31 and CR65).

3.4.5.1 Forward voltage drop.- The forward voltage drop across the diodes shall not exceed 0.8V at 1.5 ma.

3.4.5.2 Leakage current.- The leakage current through the diodes shall not exceed 0.2 ua when back-biased by 15V.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inpsections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots

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shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I

(100 percent inspection)

Characteristic	Requirement	Test Procedures
101. True-state static condition	3.4.1	4.5.2, 4.5.3.1
102. False-state, static condition	3.4.1.2	4.5.2, 4.5.3.2
103. Dynamic condition minimum clock pulse	3.4.1.3.1	4.5.3.3.1
104. Delay time comparison of input and output waveform	3.4.1.3.2 & 3.4.1.3.2.1	4.5.3.3.2
105. Head isolation diode forward voltage	3.4.3.1	4.5.5, 4.5.5.1
106. Head isolation diode leakage current	3.4.3.2	4.5.5.2
107. Test diode forward voltage drop	3.4.4.1	4.5.5.3
108. Test diode leakage current	3.4.4.2	4.5.5.4

4.3 Control test.

4.3.1 General.- One write amplifier shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - visual
302. General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the

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requirements and tests in Table I and shall then meet the requirements and tests of Table III.

TABLE III

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-Operating	3.3.4.1, & 3.4.1.3.2	4.5.3.3.2
306. Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307. Salt spray	3.3.6	4.5.1.4
308. Operating temperature	3.3.3.1, & 3.4.1.3.2	4.5.1.5, & 4.5.3.4
309. Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item	Description
1	Power supply; minus 18 volts ± 2 percent, current rating greater than or equal to 75 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from dc to 500.
2	Power supply; minus 10 volts ± 2 percent, current rating greater than or equal to 200 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from dc to 500.
3	Power supply; minus 6 volts ± 2 percent, current rating greater than or equal to 150 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from dc to 500.

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Item	Description
4	Power supply; plus 1.25 volts ± 2 percent, current rating greater than or equal to 50 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from dc to 500.
5	Power supply; plus 6 volts ± 2 percent, current rating greater than or equal to 10 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from dc to 500.
6	Power supply; plus 15 volts ± 2 percent, current rating greater than or equal to 150 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from dc to 500.
7	Power supply; plus 35 volts ± 2 percent current rating greater than or equal to 100 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from dc to 500.
8	Power supply; minus 1.5 volts ± 2 percent, current rating greater than or equal to 30 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from dc to 500.
9	Power supply; plus 50 volts ± 2 percent, current rating greater than or equal to 100 milliamps, peak-to-peak ripple less than 100 millivolts, output impedance less than 2 ohms from dc to 500.
10	Set voltage; plus 25 ± 2 volts, current rating greater than or equal to 10 milliamps, peak-to-peak ripple less than 500 millivolts, output impedance less than 5 ohms from dc to 20, connected through a 3.9K ohm ± 5 percent, 1/2 watt resistor. (See Figure 3.)
11	Reset voltage; minus 25 ± 2 volts, current rating greater than or equal to 10 milliamps, peak-to-peak ripple less than 500 millivolts, output impedance less than 5 ohms from dc to 20, connected through a 3.9K ohm ± 5 percent, 1/2 watt resistor. (See Figure 3.)
12	Oscilloscope; dual channel, rise time of scope plus associated measuring probe 50 millimicroseconds or less, sweep period 50 millimicroseconds per centimeter or greater, input resistance greater than 8 megohms input capacitance less than 25 micromicrofarads.
13	Variable amplitude clock source to supply each of the following waveforms: (a) minus 6 ± 0.5 volts for 1.8 ± .1 microseconds (true) minus 0.15 ± 0.15 volts for 0.5 ± 0.05 microseconds (false) (b) minus 3.8 ± 0.3 volts for 1.8 ± 0.1 microseconds (true) minus 0.15 ± 0.15 volt for 0.5 ± 0.05 microsecond (false) (c) minus 1.5 ± 0.1 volts for 1.8 ± 0.1 microseconds (true) minus 0.15 ± 0.15 volt for 0.5 ± 0.05 microsecond (false)

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Item	Description
	Clock rise time (10 to 90 percent amplitude) shall be 150 millimicroseconds or less. Rise time is measured during the true to false transition. Clock fall time (10 to 90 percent amplitude) shall be less than 0.4 millimicrosecond.
14	Milli-microammeter dc with a range from 0.1 microamp to 1 ampere and with an accuracy of ± 4 percent full scale deflection.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration camplitude within plus or minus 5 percent and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the write amplifier rigidly mounted in its normal operating position. Subsequent to the vibration test, the write amplifier shall show no damage or defect resulting from the test when examined by visual and tactile means. The write amplifier shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock. - This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the write amplifier shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the write amplifier in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 ± 5 percent for the 24

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hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). During the test the unit shall be operated in accordance with 4.5.3.3 to determine compliance with the requirements of 3.4.1.3 as required by 3.3.4.1.

4.5.1.4 Salt spray.- The write amplifier shall be subjected to the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the write amplifier shall be inspected for any physical damage, then tested in accordance with 4.5.3.3.2. The unit shall subsequently be inspected as specified in 4.5.1.5 prior to determining compliance with 3.4 as required by 3.3.6.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the write amplifier shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3.1 and 3.3.3.2 shall apply. While at the operating temperatures of 3.3.3.1, the write amplifier shall be tested in accordance with 4.5.3.3 to determine compliance with the requirements of 3.4.1.3 as required by 3.3.3. The temperature shall then be returned to standard temperature (60° to 90°F) and the write amplifier shall be tested in accordance with 4.5.2 to 4.5.6.2 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 write amplifiers produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance tests.- Except where otherwise specified the following test conditions shall be applied during performance tests in conjunction with the requirements of 3.4.

4.5.2.1 Power supplies.- Connect the power supplies to the respective pins as shown in Table IV.

TABLE IV - Power Supply Connections

Pin	Power Supply
8A, 32B	+15
15B	-1.5
51B	+1.25
31A	-6
41A	+6
31B	+35
36A	-10
29A	-18

4.5.2.1.1 Output lines.- Each of the output and output' lines shall be returned to the plus 15-volt power supply. The applicable pins are: 8A, 23A, 32B, 36B and 64B.

4.5.2.1.2 Power ground connections.- The dc returns of all of the power supplies shall be connected to pin 33A. This will energize the module for the remainder of the following tests as applicable and all tests shall be in conformance with the conditions of 3.4 as specified in 4.5.2.

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4.5.3 Write Amplifier I.- The following tests shall be conducted to determine compliance with 3.4.1.

4.5.3.1 True state tests.- The requirements of 3.4.1.1 shall be determined by the following procedures when momentarily applying the set-voltage (Figure 3) to pin 13B:

(a) The voltage from pin 17A to pin 52B shall be minus 0.15 ± 0.15 volts.

(b) The voltage from pin 13B to pin 52B shall be plus 0.4 volt or greater.

(c) The voltage at pin 11B shall be 0.0 ± 0.75 volt.

(d) The voltage at pin 1B shall be no more positive than plus 12 volts.

(e) The voltage at pin 25B shall be minus 7 ± 1 volts.

(f) The voltage at pin 22B shall be minus 6 ± 1 volts.

(g) The current through the output line shall be 64 ± 2 milliamps. (See Figure 1.)

(h) The current through the output' line shall be no greater than 10 microamps. (See Figure 1.)

4.5.3.2 False state test.- The requirements of 3.4.1.2 shall be determined by the following procedure when momentarily applying the reset voltage (Figure 3) to pin 13B:

(a) The voltage from pin 12B to pin 52B shall be minus 0.15 ± 0.15 volt.

(b) The voltage from pin 15A to pin 52B shall be plus 0.4 volt or greater.

(c) The voltage at pin 22B shall be 0.0 ± 0.75 volt.

(d) The voltage at pin 18B shall be no more positive than plus 12 volts.

(e) The voltage at pin 4B shall be minus 7 ± 1 volts.

(f) The voltage at pin 11B shall be minus 6 ± 1 volts.

(g) The current through the output' line shall be 64 ± 2 milliamps. (See Figure 1.)

(h) The current through the output line shall be no greater than 10 microamps. (See Figure 1.)

4.5.3.3 Dynamic tests.

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4.5.3.3.1 Minimum clock pulse tests.- The requirements of 3.4.1.3.1 shall be determined by connecting the Write Amplifier into a self-gating system as described in Figure 1. Connect a jumper from pin 34B to pin 29B. Conformance with 3.4.1.3.1.1 and 3.4.1.3.1.2 shall be determined as follows.

4.5.3.3.1.1 Triggered by 3.8V clock signal.- Connect the 3.8 volt clock signal (item 12B) to pin 29B. The waveforms at pins 2B and 24B shall each be observed. These waveforms shall be squarewaves varying between the levels of 0.5 ± 0.75 volt and minus 6 ± 0.5 volts (Figure 2).

4.5.3.3.1.2 Triggered by 1.5V clock signal.- Remove the 3.8 volt clock from pin 29B. Apply the 1.5-volt clock (item 12C) to pin 29B. The waveforms at pin 2B and 24B shall each be observed. The levels at these pins shall not change in response to the clock triggering.

4.5.3.3.2 Delay time tests.- To determine conformance with 3.4.1.3.2, repeat the connections outlined in 4.5.3.3.1. Connect the 6 volt clock signal (item 12A) to pin 29B. The waveform and delay time requirements of 3.4.1.3.2 (a) and (b) shall be determined as follows.

4.5.3.3.2.1 Compare the clock signal and the waveform (Figure 2) at pin 2B, utilizing a dual channel oscilloscope externally synchronized, to the clock signal (item 11). The delay time measured from the time a clock pulse reaches 10 percent of its total excursion (going from minus 6 ± 1 volt to minus 0.15 ± 0.15 volt) until the waveform at pin 2B reaches 90 percent of its total excursion (going from minus 0.5 ± 0.75 volt to minus 6 ± 0.5 volts) shall be less than 0.4 microsecond. The delay time measured from the time a clock pulse reaches 10 percent of its total excursion (going from minus 6 ± 1 volt to minus 0.15 ± 0.15 volt) shall be less than 0.4 microsecond. The waveform at pin 24B shall also be compared with the clock signal.

4.5.4 Write Amplifier 2.- All tests for Write Amplifier 1 specified under 4.5.3 shall be repeated for Write Amplifier 2 to determine compliance with 3.4.2. Corresponding pin connection numbers are shown in Table V for those pins which are not identical. The subordinate tests of 4.5.3 for Write Amplifier 1 are shown in the left column of Table V and the correct corresponding pin numbers for Write Amplifier 2 are shown in the right column. The following pins are identical to both amplifiers and therefore are not included in Table V: 31A, 33A, 41A, 15B, 29B, 31B, 34B, 51B, 52B.

TABLE V

Write Amplifier 1	Write Amplifier 2
Pin 8A	Pin 36B
15A	54A
17A	52A
29A	36A
1B	43B
2B	38B
4B	39B
11B	44B
12B	50A

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TABLE V (continued)

Write Amplifier 1	Write Amplifier 2
13B	51A
18B	57B
22B	56A
24B	59B
25B	61B
32B	64B

4.5.5 Head isolation diode tests.- To determine compliance with 3.4.3, all connections previously specified shall be removed from the module. This shall include all jumpers, voltage levels and ground connections. Only those voltage levels specified in the following tests shall be applied to the module.

4.5.5.1 Forward voltage.- Forward voltage shall be tested to determine compliance with 3.4.3.1 by connecting a 200 ohm ± 5%, 2 watt resistor to pin 3B and pin 63B. Return the other side of each resistor to the minus 18 volt supply, ground each of the pins specified in Table VI in sequence while reading the voltage from pin 36B to ground. This voltage shall be minus 0.5 ± 0.5 volt.

TABLE VI

Sequence No.	Pin
1	63A
2	57A
3	65B
4	61A
5	64A
6	60A
7	65A
8	59A

4.5.5.1.1 Ground each of the pins specified in Table VII in sequence while reading the voltage from pin 3B to ground. This voltage shall be minus 0.5 ± 0.5 volt.

TABLE VII

Sequence No.	Pin
1	4A
2	12A
3	3A
4	6A
5	2A
6	7A
7	1A
8	7B

4.5.5.2 Leakage current.- Leakage current shall be tested to determine compliance with 3.4.3.2 by removing the resistors that were used in 4.5.5.1 and

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connecting a 10K ohm ± 5%. 1/2 watt resistor to pin 63B and pin 3B. Return the other side of the resistors to the plus 15 volt supply. Ground each of the pins specified in Table VIII in sequence while reading the voltage from pin 63B to the plus 15 volt supply. This voltage shall not exceed minus 100 millivolts.

TABLE VIII

Sequence No.	Pin
1	63A
2	57A
3	65B
4	61A
5	64A
6	60A
7	65A
8	59A

4.5.5.2.1 Ground each of the pins specified in Table IX in sequence while reading the voltage from pin 3B to the plus 15 volt supply. This voltage shall not exceed minus 100 millivolts.

TABLE IX

Sequence No.	Pin
1	4A
2	12A
3	3A
4	6A
5	2A
6	7A
7	1A
8	7B

4.5.6 Test diodes tests.

4.5.6.1 Amplifier 1.

4.5.6.1.1 Forward voltage drop.- The forward voltage drop shall be tested to determine compliance with 3.4.4.1 by connecting the minus 18 volt power supply to pin 50B. Voltage drop shall be determined by the following procedures when grounded as specified:

(a) Ground pin 22A. The voltage at pins 10B, 8B, 19B and 20B shall each be minus 0.6 volt. Remove the ground from pin 22A.

(b) Ground pin 20A. The voltage at pins 10B, 8B, 19B and 20B shall each be minus 0.6 ± 0.6 volt. Remove the ground from pin 20A.

(c) Ground pin 26B. The voltage at pin 20B shall be minus 0.5 ± 0.5 volt. Remove the ground from pin 26B.

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(d) Ground pin 33B. The voltage at pin 8B shall be minus 0.5 ± 0.5 volt. Remove the ground from pin 33B.

(e) Ground pin 2B. The voltage at pin 8B shall be minus 0.5 ± 0.5 volt. Remove the ground from pin 2B.

(f) Ground pin 23B. The voltage at pin 20B shall be minus 0.5 ± 0.5 volt. Remove the ground from pin 23B. Upon completion of the forward voltage drop tests remove the minus 18-volt power supply from pin 50B and all leads.

4.5.6.1.2 Leakage current.- Leakage current shall be tested to determine compliance with 3.4.4.2 by connecting the plus 15 volt power supply to pin 49B. Leakage shall be determined by the following procedures when grounded as specified:

(a) Ground pins 33B and 26B. The voltage from the plus 15-volt power supply to pins 8B and 20B shall be less than 0.050 volt. Remove the ground from pins 33B and 26B.

(b) Ground pins 2B and 23B. The voltage from the plus 15-volt power supply to pins 8B and 20B shall be less than 1.7 volts. Remove the ground from pins 2B and 23B.

(c) Ground pin 14A. The voltage from the plus 15-volt power supply to pins 8B and 20B shall be less than 1.7 volts. Remove the ground from pin 14A and after completion of leakage current tests remove the plus 15-volt power supply from pin 49B.

4.5.6.2 Write Amplifier 2 test diodes.- The tests for the test diodes for Write Amplifier 1 specified under 4.5.6.1 shall be repeated for Amplifier 2. Corresponding pin connection numbers are shown in Table X for those pins which are not identical. The pins specified in the subordinate tests of 4.5.6.1 for Write Amplifier 1 are shown in the left column of Table X and the correct corresponding pin numbers for Write Amplifier 2 are shown in the right column. The following pins are identical to both amplifiers and therefore are not included in Table X: 26B, 33B, 49B, 50B.

TABLE X

Write Amplifier 1	Write Amplifier 2
Pin 14A	Pin 54B
20A	58B
22A	58A
2B	38B
8B	45B
10B	46B
19B	56B
20B	55B
23B	62B

5. PREPARATION FOR DELIVERY

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5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The Writer Amplifier contains two bistable flip-flop circuits with emitter follower output drivers which supply regulated write currents to the memory write heads of the Gun Direction Computer: M18. The head isolation diodes on the module are used to isolate a selected write head from other write heads within the computer's write-switch matrix.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (see 5.1).
(e) A block of serial numbers to identify each write amplifier circuit board.

Custodian:

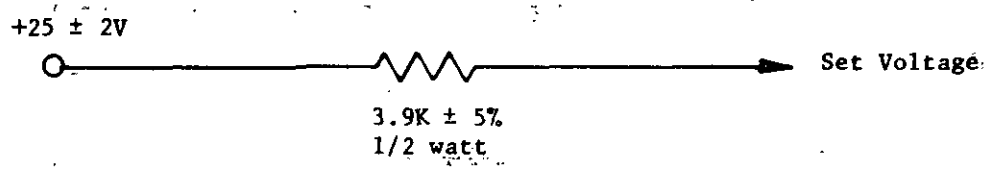
Army - MU

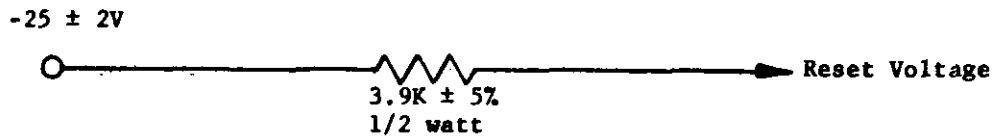
Preparing activity:

Army - MU

Project No. 1220-A190Y

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NOTE: Apply voltages only momentarily.

FIGURE 3 - Set and Reset Voltages - Write Amplifier

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEM PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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